表 1: 层间边际贡献—ΔAUROC 与统计显著性

转移	加入的组件	ΔAUROC	DeLong z	p-value	ΔCohen_d	ΔBrier
L0 → L1	+ PBC Detection	+0.4643	22.026	0.0000	+5.0077	−0.1066
L1 → L2	+ Evidence + Identity	+0.0000	0.000	1.0000 —	+1.4815	−0.0299
L2 → L3	+ Observer (Full System)	+0.0134	0.997	0.3189 —	+0.3744	−0.0021